Business Intelligence Overview

About Business Intelligence

Business Intelligence (BI) is the combination of processes and technology an organization uses to collect, prepare, analyze and present its data so that managers can make better decisions. IBM describes it as a set of processes for gathering, managing and analyzing organizational data to produce insights that guide strategy and day-to-day operations (IBM). At its core, BI is about providing historical, current, and predictive views of business operations.

BI is mainly descriptive and diagnostic; it acts as the foundation for planning as it tells management what happened and why. Typically, it involves five stages:

1. Data Sources: transactional systems such as sales, inventory and CRM records
2. Data Preparation: extracting, inspecting, cleaning and transforming the data
3. Modelling and Analysis: defining relationships and calculations so the data answers business questions
4. Visualization: presenting results in a dashboard using charts, maps, tables, and other visuals
5. Action: turning findings into decisions and tracking them against KPIs

The Importance of Dashboards

A dashboard is a visual representation of the most important measures of a business, arranged so they can be read at a glance. Organizations use dashboards for:

- Data Consolidation: it brings business data (sales, customer and product) from many systems into one view
- Accessibility: raw data is often difficult for non-technical stakeholders to interpret. Dashboards make complex data accessible to everyone.
- Goal Alignment: by displaying core KPIs, dashboards ensure that all team members are aligned and focused on the company's primary objectives.
- Performance Monitoring: business KPIs show whether the business is on track
- It shortens the time spent on acquiring information on business performance

How Dashboards Improve Decision-Making

Dashboards make decision-making faster, grounds decisions in evidence, and makes it easier to check them afterwards.

- Speed: team members can get updated information the day the data refreshes, not weeks later in a month-end report.
- Evidence-Backed Decisions: decisions on pricing, discounts or where to expand rest on actual sales and profit figures.
- Drill-Down to Causes: slicers and filters allow users to **narrow down a** company-wide total to a region, a category, or a single product to find what is driving a result.
- Spotting Patterns: trend lines and comparisons reveal seasonality, growth and loss-making areas that are invisible in raw tables.
- Accountability: ongoing monitoring confirms whether a past decision worked, so poor choices are corrected rather than repeated (FrankCrum).
- Testing Options: comparing segments, channels or regions side by side shows where effort and money produce the best return.

Business Objective

The objective is to give the retailer's senior management one interactive view of sales, profit, customers and regions, so they can see where the business makes and loses money and act on it. The dataset chosen is the **Superstore Sales Dataset** (Option 1): 9,993 order lines (after removing one duplicate) from 5,009 orders placed by 793 customers across the United States between January 2014 and December 2017.

Over the four years the company recorded about $2.30 million in sales and $286,000 in profit, a 12.5% profit margin.

The dashboard is designed to answer the seven business questions in the brief.